Clustering Analysis

Objective

The goal of this stage is to identify distinct customer segments within the dataset using unsupervised machine learning. By clustering similar customers, the business can:

- Understand behavioral patterns influencing churn
- Identify high-risk or high-value segments
- Provide actionable insights for retention strategies

Three deliverables were completed:

- Determine the optimal number of clusters (Elbow Method)
- Train a K-Means clustering model
- Visualize clusters and generate cluster profiles for interpretation

Dataset Used

The clustering analysis used the scaled training dataset (X_train_scaled), which was preprocessed for missing values, categorical encoding, and scaling.

Attribute	Value
Dataset Name	X_train_scaled
Number of Observations	5392
Number of Features	11

```
<class 'pandas.DataFrame'>
RangeIndex: 5392 entries, 0 to 5391
Data columns (total 11 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   SeniorCitizen                5392 non-null   float64
 1   tenure                       5392 non-null   float64
 2   MonthlyCharges               5392 non-null   float64
 3   gender_Male                  5392 non-null   float64
 4   Dependents_Yes               5392 non-null   float64
 5   PhoneService_Yes             5392 non-null   float64
 6   MultipleLines_Yes            5392 non-null   float64
 7   InternetService_Fiber optic  5392 non-null   float64
 8   Contract_One year            5392 non-null   float64
 9   Contract_Two year            5392 non-null   float64
 10  Churn_Yes                    5392 non-null   float64
dtypes: float64(11)
```

Elbow Method Approach

The elbow method works by computing the Within-Cluster Sum of Squares (WCSS) for different values of K.

WCSS measures the total squared distance between each data point and the centroid of its assigned cluster.

A lower WCSS indicates that data points are closer to their cluster center, meaning more compact clusters.

The algorithm was executed for cluster values ranging from:

K = 1 to K = 10

For each value of K:

1. A K-Means model was trained.
2. WCSS was calculated.
3. The value was stored for comparison.

WCSS Results

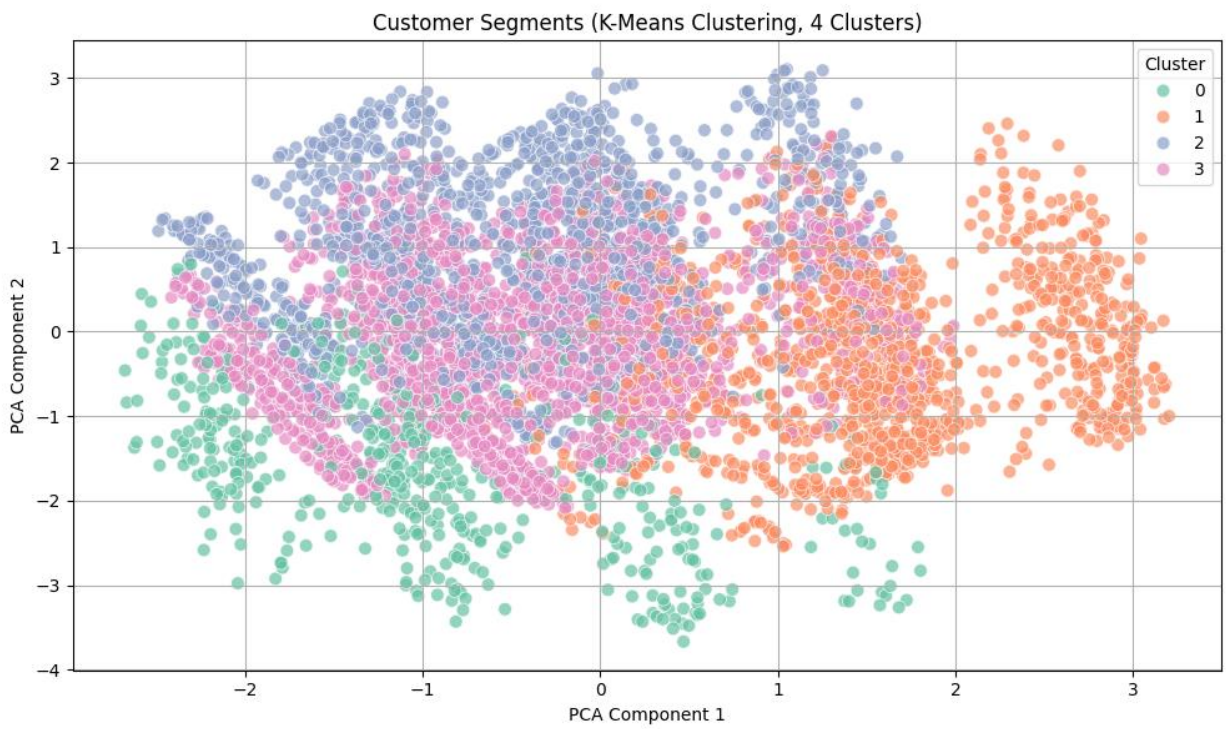
The following WCSS values were obtained during the clustering experiments:

```
Calculating WCSS for different cluster values...
Clusters: 1, WCSS: 59312.0
Clusters: 2, WCSS: 53512.21510848815
Clusters: 3, WCSS: 47701.63833472455
Clusters: 4, WCSS: 43489.06482342623
Clusters: 5, WCSS: 40024.56937165238
Clusters: 6, WCSS: 38237.58158804224
Clusters: 7, WCSS: 37202.11318617924
Clusters: 8, WCSS: 34366.88541418472
Clusters: 9, WCSS: 33287.864344339505
Clusters: 10, WCSS: 32396.91533706331
```

Elbow Curve Visualization

To visually determine the optimal number of clusters, the WCSS values were plotted against the number of clusters.

The elbow point represents the value of K where the reduction in WCSS begins to slow significantly.



The plot shows the relationship between the number of clusters and the WCSS metric.

Interpretation of Results

From the elbow curve, it can be observed that:

- The WCSS decreases rapidly between K = 1 and K = 4
- After K = 4 or K = 5, the decrease becomes more gradual

This suggests that adding more clusters beyond this point provides diminishing returns in clustering quality.

Therefore, the elbow point appears around:

K ≈ 4 or K ≈ 5

These values represent a reasonable balance between:

- cluster compactness
- model simplicity.

Selected Cluster Count

Based on the elbow method analysis, the project will proceed with:

K = 4 clusters

This value provides a good balance between minimizing WCSS and maintaining interpretable customer segments.

The next stage will involve training the K-Means clustering model using K = 4.

Training the K-Means Model

The K-Means algorithm was initialized with the following parameters:

- n_clusters = 4 (based on elbow method)
- random_state = 42 (ensures reproducibility)
- n_init = 10 (number of initializations)

The model was trained on the scaled features, and cluster labels were assigned to each customer.

Cluster Centers:

The centroid of each cluster represents the average feature values for customers in that segment.

```
Cluster Centers:

   SeniorCitizen    tenure  MonthlyCharges  gender_Male  Dependents_Yes  \
0      -0.056123 -0.059626       -0.805599     0.056922       -0.038801
1       0.258596 -0.611406        0.441107    -0.066484       -0.274304
2      -0.076147  0.243301       -0.026683     0.010784        0.109796
3      -0.090475  0.218731       -0.037848     0.018269        0.100952

   PhoneService_Yes  MultipleLines_Yes  InternetService_Fiber optic  \
0         -2.952171          -0.089703                    -0.002683
1          0.338734           0.002556                    -0.019105
2          0.338734           1.164308                     0.438684
3          0.338734          -0.858879                    -0.319446

   Contract_One year  Contract_Two year  Churn_Yes
0           0.018990           0.015795  -0.038746
1          -0.037555          -0.018562   1.662148
2           0.006667           0.104745  -0.583911
3           0.013493          -0.071927  -0.598276
```

Cluster Assignment

Each customer was assigned a cluster label (0–3). The first 20 assigned labels are:

```
Cluster labels assigned successfully.
First 20 cluster labels:
 [2 3 3 3 3 3 0 3 0 2 2 3 1 1 2 1 1 1 0 3]
```

Cluster Distribution

The number of customers in each cluster is:

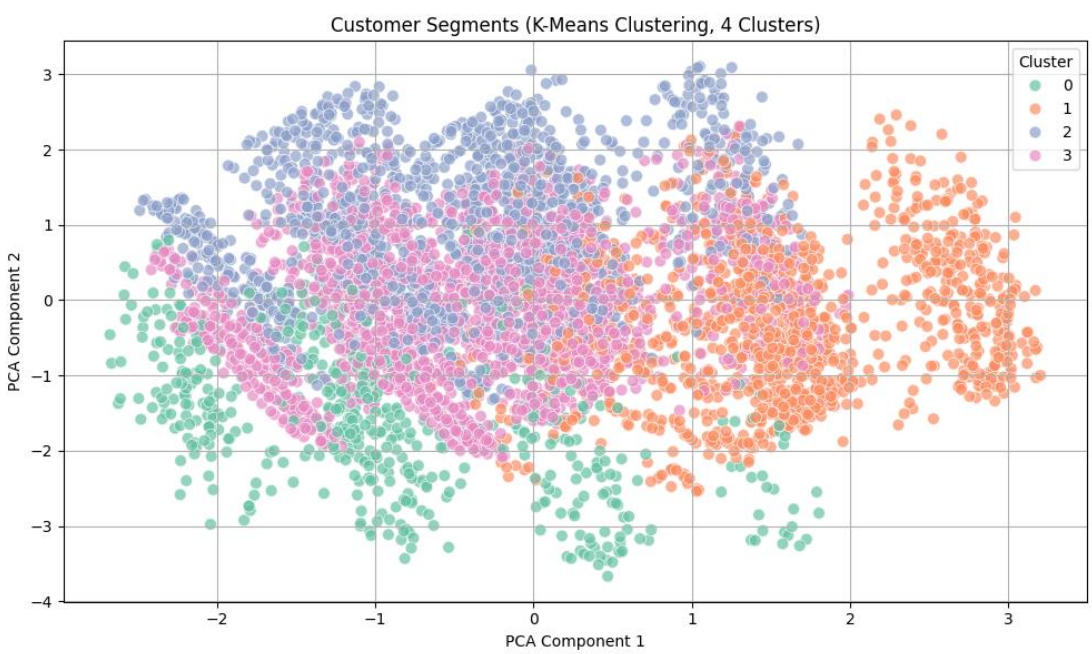
```
Cluster column added.

Cluster Distribution:
Cluster
3    2024
2    1533
1    1280
0     555
Name: count, dtype: int64
```

Interpretation:

- **Cluster 3** is the largest segment (2024 customers)
- **Cluster 0** is the smallest segment (555 customers)
- This shows clear differentiation in customer grouping based on features

Visualizations

- **PCA 2D plot** to show customer clusters in a 2D projection



Summary

The K-Means clustering model was successfully trained with 4 clusters, producing:

- Cluster centroids
- Cluster labels for each customer
- Clustered dataset for further visualization and business insights

These outputs will be used in the next stage to create visualizations, interpret segments, and derive actionable insights for churn prediction.